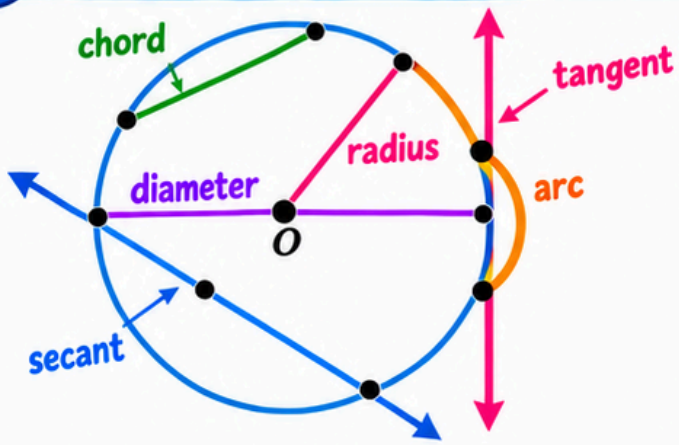
Geometry Circles Cheat Sheet

Unit Focus: *parts of a circle, circumference, area, arcs, sectors, and angles*

1 KEY VOCABULARY

center	The fixed point in the middle of a circle.
radius	A segment from the center to a point on the circle.
diameter	A segment that passes through the center with endpoints on the circle (length is $2r$).
chord	A segment with both endpoints on the circle.
tangent	A line that touches the circle at exactly one point.
secant	A line that intersects the circle at two points.
arc	A part of the circumference between two points.
central angle	An angle with vertex at the center of the circle that intercepts an arc.
inscribed angle	An angle with vertex on the circle that intercepts an arc.

2 PARTS OF A CIRCLE



3 CIRCUMFERENCE OF A CIRCLE

The **circumference** is the distance around a circle.

$$C = 2\pi r \quad \text{or} \quad C = \pi d$$



Example: Find the circumference of a circle with radius $r = 5$.

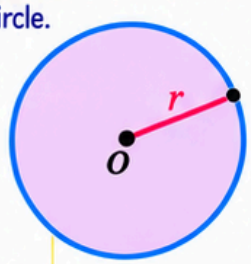
$C = 2\pi(5) = 10\pi$

$C \approx 31.4$ units (using $\pi \approx 3.14$)

4 AREA OF A CIRCLE

The **area** is the amount of space inside a circle.

$$A = \pi r^2$$



Example: Find the area of a circle with radius $r = 6$.

$A = \pi(6)^2 = 36\pi$

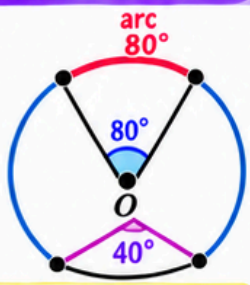
$A \approx 113.1$ square units (using $\pi \approx 3.14$)

5 CENTRAL AND INSCRIBED ANGLES

A **central angle** has its vertex at the center.
An **inscribed angle** has its vertex on the circle.

$m\angle(central) =$ measure of its intercepted arc

$m\angle(inscribed) = \frac{1}{2}$ (measure of its intercepted arc)

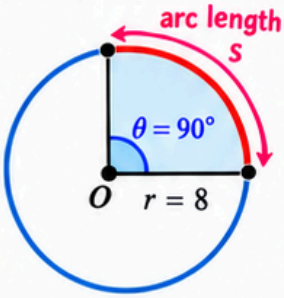


Example: If the intercepted arc measures 80°, then the inscribed angle measures $\frac{1}{2}(80°) = 40°$.

6 ARC LENGTH

The **arc length** is the distance along a portion of the circumference.

$$s = \frac{\theta}{360°} \cdot 2\pi r$$



Example: Find the arc length for a central angle of 90° in a circle with radius $r = 8$.

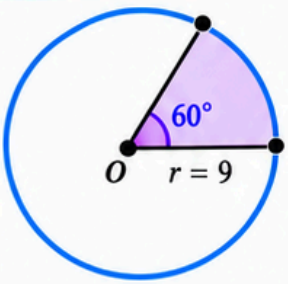
$s = \frac{90°}{360°} \cdot 2\pi(8) = \frac{1}{4} \cdot 16\pi = 4\pi$

$s = 4\pi$ units

7 SECTOR AREA

A **sector** is the region inside a circle bounded by two radii and an arc.

$$A_{sector} = \frac{\theta}{360°} \cdot \pi r^2$$



Example: Find the area of a sector with central angle 60° and radius $r = 9$.

$A_{sector} = \frac{60°}{360°} \cdot \pi(9)^2 = \frac{1}{6} \cdot 81\pi = 13.5\pi$

$A_{sector} = 13.5\pi$ square units

8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- The radius is half the diameter ($d = 2r$).
- A tangent is perpendicular to the radius at the point of tangency (90°).
- Use degrees (not radians) in the arc length and sector area formulas.
- A central angle's measure equals the measure of its intercepted arc.
- An inscribed angle's measure is half the measure of its intercepted arc.

COMMON MISTAKES:

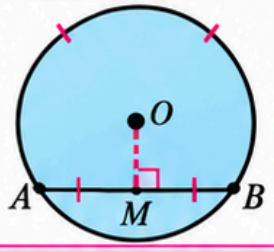
- Confusing the radius and diameter.
- Thinking a chord is a tangent (a chord has two endpoints on the circle).
- Forgetting to use degrees in the arc length and sector area formulas.
- Mixing up circumference and area (C measures distance around, A measures area inside).
- Assuming inscribed angles are equal to the intercepted arc (they are half).

Geometry Circles Cheat Sheet

More Practice: *circle angles, tangents, and applications*

1 CHORD RELATIONSHIPS

- Congruent chords intercept congruent arcs.
- Congruent arcs have congruent chords.
- A radius or diameter perpendicular to a chord bisects the chord and its intercepted arc.

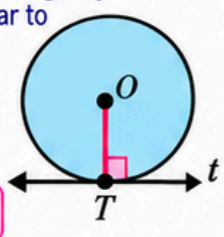


If $\overline{AB}$ is a chord and $\overline{OM} \perp \overline{AB}$, then $AM = MB$ and $\overset{\frown}{AB}$ is bisected.

2 TANGENTS

Tangent Line

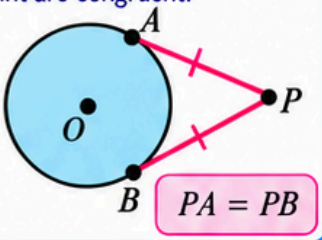
A tangent touches a circle at exactly one point. A radius to the point of tangency is perpendicular to the tangent.



$\overline{OT} \perp t$

Tangent Segments from the Same Point

Two tangent segments from the same external point are congruent.



$PA = PB$

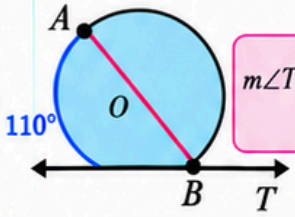
3 ANGLE FORMED BY A TANGENT AND A CHORD

An angle formed by a tangent and a chord has a measure equal to half the measure of its intercepted arc.

$$m\angle = \frac{1}{2}(\text{intercepted arc})$$

Example:

Find the measure of $\angle TBA$ if $\overset{\frown}{AB} = 110°$.



$$m\angle TBA = \frac{1}{2}(110°) = 55°$$

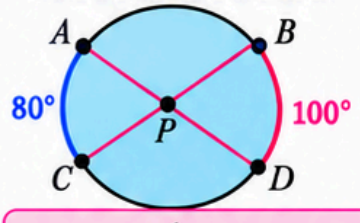
4 ANGLES FORMED INSIDE THE CIRCLE

An angle formed by two chords that intersect inside a circle has a measure equal to half the sum of the measures of the intercepted arcs.

$$m\angle = \frac{1}{2}(\text{arc}_1 + \text{arc}_2)$$

Example:

Find the measure of $\angle APB$.



$$m\angle APB = \frac{1}{2}(80° + 100°) = 90°$$

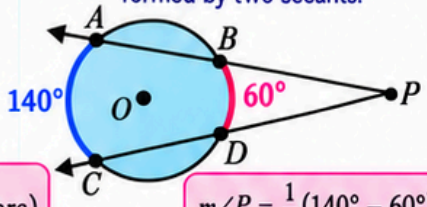
5 ANGLES FORMED OUTSIDE THE CIRCLE

An angle formed by two secants, a secant and a tangent, or two tangents with a common external point has a measure equal to half the difference of the measures of the intercepted arcs.

$$m\angle = \frac{1}{2}(\text{larger arc} - \text{smaller arc})$$

Example:

Find the measure of $\angle P$ formed by two secants.



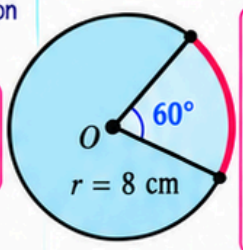
$$m\angle P = \frac{1}{2}(140° - 60°) = 40°$$

6 ARC LENGTH EXAMPLE

The length of an arc is a fraction of the circumference based on its central angle.

$$s = \frac{\theta}{360°} \cdot 2\pi r$$

Example: Find the arc length s.



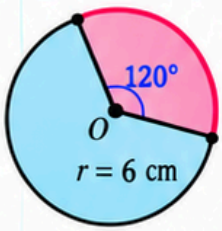
$$s = \frac{60°}{360°} \cdot 2\pi(8) = \frac{1}{6} \cdot 16\pi = \frac{8\pi}{3} \text{ cm}$$

7 SECTOR AREA EXAMPLE

The area of a sector is a fraction of the area of the circle based on its central angle.

$$A_{sector} = \frac{\theta}{360°} \cdot \pi r^2$$

Example: Find the area of the sector.



$$A = \frac{120°}{360°} \cdot \pi(6)^2 = \frac{1}{3} \cdot 36\pi = 12\pi \text{ cm}^2$$

8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Congruent chords ⟷ congruent arcs (and vice versa).
- A radius is perpendicular to a tangent at the point of tangency.
- A radius/diameter perpendicular to a chord bisects the chord and arc.
- Angles formed inside the circle use the **SUM** of the intercepted arcs.
- Angles formed outside the circle use the **DIFFERENCE** of the intercepted arcs.
- Inscribed angles and tangent-chord angles are HALF the measure of the intercepted arc.

COMMON MISTAKES:

- Using the wrong arc (not the intercepted arc).
- Forgetting a tangent touches at only one point.
- Thinking a radius along a chord bisects it (it must be perpendicular).
- Using the difference instead of the sum for inside angles (or sum instead of difference for outside angles).
- Forgetting to divide by 2 for inscribed or tangent-chord angles.